Handify

PROJECT OVERVIEW

Title - Handify

Tagline - Bridging Communication for the Differently-Abled

Team Members :

Vidit Sharma

Vansh Sihania

Saumya

Sumit Chauhan

Mentor - Ms. Anjali Chauhan

PROBLEM STATEMENT

Main Challenge :

How can we enable easier communication for differently-abled individuals, especially those who rely on sign language.

Current Issue :

Most people are unfamiliar with sign language, creating a communication barrier for the deaf and hard of hearing community.

PROPOSED SOLUTION

Handify Aims to

Detect hand gestures used in sign language.

Convert those gestures into text.

Help differently-abled individuals communicate effortlessly with others.

TECHNOLOGIES USED

Computer Vision : For Frame capturing, processing, color conversion etc

Python : Programming language for building the core logic.

Machine Learning : Recognizing patterns in gestures.

Mediapipe : Hand detection and landmark extraction.

HOW IT WORKS

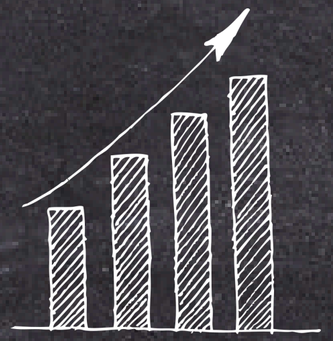
1. **Hand Gesture Detection :**
   - Using a camera (Webcam/Phone camera).
   - Detect the hand movements and capture the gesture.
2. **Processing :**
   - OpenCV processes the gesture .
   - Random Forest classifier identifies the specific sign..
3. **Conversion :**
   - The recognized sign is converted into text .

FUTURE SCOPE

Improve accuracy of gesture recognition.

Expand to different languages and gestures.

Create a mobile app for easier accessibility.



CURRENT STATUS

- ✅ Implementation is done
- ✅ Webcam integration done using OpenCV
- ✅ Hand tracking implemented using MediaPipe
- ✅ Research paper is completed and published
- ✅ Project Report is completed

CONCLUSION

Handify is a **powerful solution** designed **to assist** differently-abled individuals by translating hand gestures into text. With cutting-edge technologies like OpenCV, machine learning, and Mediapipe, the project aims to make communication more accessible and inclusive. As we move forward, this system has the potential to significantly improve the lives of those who rely on sign language for communication.

THANK YOU